DBMS UNIT 4

UNIT – IV Question Bank Categorized by Topic

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1. Informal Design Guidelines for Relation Schemas

Q1 (6 Marks)

Discuss the informal design guidelines for relational schema in detail.

Q2 (8 Marks)

List and explain the informal guidelines for relation schema design.

Q3 (10 Marks)

State the informal guidelines for relational schema design. Illustrate how violation of these guidelines may be harmful.

Q4 (7 Marks)

Explain with an example the Informal guidelines for a relation schemas.

Q5 (5 Marks)

Explain any two informal design guidelines for relation schemas.

Q6 (6 Marks)

Briefly discuss different Informal Design Guidelines for Relational Databases.

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2. Functional Dependencies

A. Basic Functional Dependency Concepts

Q7 (8 Marks)

Define Functional dependency. Discuss the inference rules for Functional dependency.

Q8 (6 Marks)

Define fully functional dependency. Explain the 2 NF with an example.

Q9 (5 Marks)

Define fully functional dependency. Explain the 2 NF with an example.

Q10 (6 Marks)

Define MVD and explain 4NF related to it with an example.

B. Checking Functional Dependencies

Q11 (6 Marks)

Given the relation r(R). State whether the following functional dependencies are satisfied by the relation or not.

i) $A \rightarrow B$

ii) $AB \rightarrow C$

iii) $C \rightarrow A$

iv) $BC \rightarrow A$

A	B	C
1	4	2
3	4	6
3	4	6
7	3	8
9	1	0

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3. Inference Rules (Armstrong’s Axioms)

Q12 (6 Marks)

Prove the following inference rules.

- IR4 (decomposition, or projective, rule): $\{X \rightarrow YZ\} \models X \rightarrow Y$.
- IR5 (union, or additive, rule): $\{X \rightarrow Y, X \rightarrow Z\} \models X \rightarrow YZ$.
- IR6 (pseudotransitive rule): $\{X \rightarrow Y, WY \rightarrow Z\} \models WX \rightarrow Z$.

Q13 (4 Marks)

Prove the following Inference Rules for functional dependencies

i. $\{X \rightarrow Y, XY \rightarrow Z\} \models \{X \rightarrow Z\}$

ii. $IX \rightarrow Y, Z \rightarrow W\} \models \{XZ \rightarrow YW\}$

Q14 (6 Marks)

Briefly discuss Armstrong’s Inference Rules for Functional Dependency.

Q15 (6 Marks)

Let R(A,B,C,D,E,G,H) be a relation schema with set of functional dependencies:

$F = \{AB \rightarrow C, B \rightarrow D, CD \rightarrow E, CE \rightarrow GH, G \rightarrow A\}$.

With the help of inference rules prove the following:

i. Exhibit a derivation of $AB \rightarrow E$ from F.

ii. Exhibit a derivation of $GB \rightarrow C$ from F.

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4. Equivalence of Functional Dependency Sets

Q16 (9 Marks)

Write an algorithm to find the closure of given functional dependency.

Given the following set of functional dependency check whether they are equivalent or not.

Relation R = (A, B, C, D, E, F)

F1 = { A → C, AC → D, E → AD, E → F }

F2 = { A → CD, E → AF }

Q17 (6 Marks)

Assume a relation R is defined over the attributes R(A,B,C,D,E,F) and the functional dependencies are

F = {A → C, AC → D, E → AD, E → F}

and G = {A → CD, E → AF}.

Check whether G and H are equivalent or not.

Q18 (6 Marks)

Assume a relation R is defined over the attributes R(A,B,C,D,E,F) and the functional dependencies are

F = {A → C, AC → D, E → AD, E → F} and G = {A → CD, E → AF}.

Check whether G and H are equivalent or not.

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5. Minimal Cover / Canonical Cover

Q19 (8 Marks)

A set of FD's of a relation R(W,X,Y,Z) is

{X → YZ, Y → Z, X → Y, XY → Z, XZ → W}.

Find the minimal cover for this set of Functional Dependencies.

Q20 (5 Marks)

Explain minimal cover of set of functional dependency. Write an algorithm to find the minimal cover set of the functional dependency.

Q21 (8 Marks)

A set of FD’s for the relation {A, B, C, D} F:{B → A, D → A, AB → D}.

Find a minimum cover for this set of FD’s illustrating all the inferences used.

Q22 (6 Marks)

Given the set of FD’s,

F:{ A → BC, B → CE, A → E, AC → H, D → B}.

Find the minimal cover of F.

Q23 (9 Marks)

A set of FD's for the relation R{A,B,C,D,E,F} is

A → BC,

C → A,

BC → D,

AC → D,

B → C,

EC → FA,

CF → BD,

D → E

Find a minimum cover for this set of FD'S.

Q24 (5 Marks)

A set of FD’s for the relation R{A,B,C,D,E,F} is

AB → C, C → A, BC → D, ACD → B, BE → C, EC → FA, CF → BD, D → E.

Find a minimum cover for this set of FD’s.

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6. First Normal Form (1NF)

Q25 (8 Marks)

Why Normalization is Important? When you say that the relation schema satisfies First Normal form Give an example.

Q26 (7 Marks)

If the relation contains multi valued attribute what are different techniques to achieve 1NF. Illustrate with an example and also find which technique is the best. Justify your answer.

Q27 (8 Marks)

What is the need for normalization? Explain 1NF, 2NF, 3NF with examples.

Q28 (10 Marks)

What is the need for normalization? Explain 1NF, 2NF, 3NF with examples.

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7. Second Normal Form (2NF)

Q29 (5 Marks)

Explain the 2NF with an example.

Q30 (7 Marks)

What is the need for Normalization? Explain second normal form.

Consider the relation EMP-PROJ = {SSN, Pnumber, Hours, Ename, Pname, Plocation}.

Assume {SSN, Pnumber} as primary key. The dependencies are:

SSN Pnumber → {Hours}

SSN → {Ename}

Pnumber → {Pname, Plocation}

Normalize the above relation into 2NF.

Q31 (6 Marks)

Given a relation R(A, B, C, D) and Functional Dependency set

FD = { AB → CD, B → C }, determine whether the given R is in 2NF. If not convert it into 2NF.

Q32 (6 Marks)

Given a relation R(A, B, C, D) and Functional Dependency set

FD = { AB → CD, B → C }, determine whether the given R is in 2NF. If not convert it into 2NF.

Q33 (6 Marks)

Explain fully functional dependency. Explain the 2 NF with an example.

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8. Third Normal Form (3NF)

Q34 (6 Marks)

Define transitive dependency. Explain 3NF with an example.

Q35 (9 Marks)

Differentiate between 3NF and BCNF. For the given relation Patient:

Patient (Patno, PatName, appNo, time, doctor)

FD's:

{Patno → PatName,

Patno → Time, doctor

Time → appNo

appNo → Patno}

i. Find the keys of the relation.

ii. Check whether the relation is in 3NF and BCNF. If not decompose the relation into 3NF and BCNF.

Q36 (10 Marks)

Consider the universal relation

R={A,B,C,D,E,F,G,H,I,J} and the set of functional dependencies

F={{A,B}→{C}, {A}→{D,E}, {B}→{F}, {F}→{G,H}, {D}→{I,J}}

What is the key for R?

Decompose R into 2 NF and then into 3NF relations.

Q37 (8 Marks)

Consider the universal relation

R={A,B,C,D,E,F,G,H,I,J} and the set of functional dependencies

F={{A,B}→{C}, {A}→{D,E}, {B}→{F}, {F}→{G,H}, {D}→{I,J}}

What is the key for R?

Decompose R into 2 NF and then into 3NF relations.

Q38 (8 Marks)

Explain normal forms based on primary keys and super keys and functional dependencies with examples.

MISSED-Explain the general guidelines for a relation in second and third normal form with an example for each.” (06)

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9. Boyce Codd Normal Form (BCNF)

Q39 (6 Marks)

Define BCNF. How does it differ from 3NF? Why is it considered a stronger form of 3NF?

Q40 (7 Marks)

Explain BCNF. State if the following relation is into BCNF. If not, decompose the same into BCNF. State the candidate key.

STUDENT (STUDENT, COURSE, INSTRUCTOR)

FD1: {Student, Course} → Instructor

FD2: Instructor → Course

Q41 (6 Marks)

Consider the relation R with four attributes A,B,C,D with the following sets of FD's

C → D, C → A, B → C

i. Identify the candidate key(s) for R.

ii. Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF).

iii. If R is not in BCNF, decompose it into a set of BCNF relations.

Q42 (9 Marks)

Consider a relation with schema R(A, B, C, D) and FD's {AB→C, C→D, D→A}.

i. Indicate the nontrivial FD's that can be inferred from the given FD's?

ii. What are all candidate keys of R?

iii. Is this relation in BCNF? If not Decompose R into 2NF, 3NF and BCNF relations.

Q43 (8 Marks)

Determine the normal form for the following relation and justify the same.

BOOK(ISBN, Title, Publisher, Address)

FD's:

ISBN → Title

ISBN → Publisher

Publisher → Address

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10. Properties of Relational Decomposition

A. Lossless Join / Non-additive Join

Q44 (6 Marks)

Let R = ABCDE, R1 = AD, R2 = AB, R3 = BE, R4 = CDE, and R5 = AE.

Let the functional dependencies be:

A → C, B → C, C → D, DE → C, CE → A.

Apply algorithm to test if the decomposition of R into {R1,R2,R3,R4,R5} is a lossless join decomposition.

Q45 (6 Marks)

Let R = ABCDE, R1 = AD, R2 = AB, R3 = BE, R4 = CDE, and R5 = AE.

Let the functional dependencies be:

A → C, B → C, C → D, DE → C, CE → A.

Apply algorithm to test if the decomposition of R into {R1,...,R5} is a lossless join decomposition or not.

Q46 (8 Marks)

Write the algorithm for testing Lossless (nonadditive) Join Property.

Consider the schema R=ABCDE, subjected to FDs

F={A→C, B→C, C→D, DE→C, CE→A}, and the decomposed relations are {AD, AB, BE, CDE, AE}.

Check whether the decomposition is satisfying Lossless (nonadditive) Join Property.

Q47 (6 Marks)

Write the algorithm for testing of non-additive join property.

Q48 (4 Marks)

Discuss the problem of spurious tuples and how we may prevent it.

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11. Algorithms for Relational Database Schema Design

A. Closure Algorithm

Q49 (9 Marks)

Write an algorithm to find the closure of given functional dependency.

Q50 (6 Marks)

List the Inference rules for functional dependencies. Write the algorithm to determine the closure of X (set of attributes) under F (set of functional dependencies).

B. Relational Synthesis / 3NF Decomposition

Q51 (9 Marks)

A set of FD's for the relation

R{A,B,C,D,E,F} is

AB→C, C→A, BC→D, ACD→B, BE→C, EC→FA, CF→BD, D→E.

Find a minimum cover for this set of FD's.

Decompose the relation Using Relational Synthesis algorithm into 3NF.

MISSED-Explain Multi-valued Dependencies, Fourth Normal Form, Join Dependencies and Fifth Normal Form with examples. (10 Marks)

What normal form is the given relation in? Apply normalization until you cannot decompose the relations further. State the reasons behind each decomposition. (08 Marks)

Determine the normal form for the following relation and justify the same.

BOOK(ISBN, Title, Publisher, Address)

FD's:

- ISBN → Title
- ISBN → Publisher
- Publisher → Address

(08 Marks)

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12. Anomalies in Relational Design

Q52 (5 Marks)

Discuss the different types of anomalies in handling the relations with an example.

Q53 (7 Marks)

Discuss insertion, deletion and modification anomalies. Why are they considered bad? Illustrate with examples.